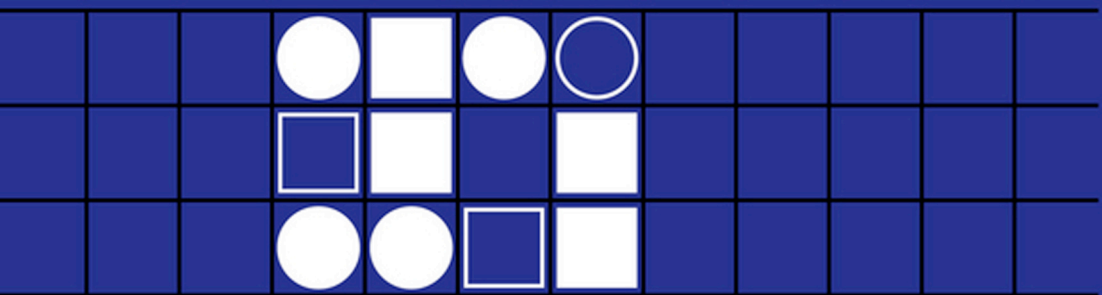


PROBABILITY & STATISTICS WITH RELIABILITY, QUEUEING, AND COMPUTER SCIENCE APPLICATIONS

SECOND EDITION



KISHOR S. TRVEDI

WILEY

**Probability and Statistics with
Reliability, Queuing, and
Computer Science Applications**

Probability and Statistics with Reliability, Queuing, and Computer Science Applications

Kishor S. Trivedi

Duke University

Durham, North Carolina

WILEY

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Preface to the Paperback Edition

Nearly 15 years have passed since the publication of the second edition of this book by John Wiley. Indian edition (of the first edition this book from 1982), published by Prentice-Hall India, is still in print and doing quite well. Asian edition (of the second edition from 2001) is published by John Wiley Asia. In 2015, a Chinese translation of the second edition has been published. I did not see the need for adding or subtracting significant amount of material to produce a new, third edition of the “bluebook” as it is popularly known. However, I took the opportunity to correct nearly 100 minor errors from the book before supplying the camera ready copy to the publisher. I have added some more terms in the subject index leading to an extra page at the end. Thus, though this is not a new edition, it is a significant new reprint. I like to remind the readers that a complete solution manual is available to instructors as well the power point slides of all chapters. Instructors may also wish to use the SHARPE software package as a pedagogic aid. (See <http://sharpe.pratt.duke.edu/> for further details.)

I wish to thank many friends for pointing out and/or help fix the errors—Javier Alonso López, Yonghuan Cao, Xiaolin Chang, Olivia Das, Jie Feng, Marcos A. M. Ferreira, Lance Fiondella, Ravi Iyer, Michael Kwok, Bharat Madan, José M. Martínez, Rivalino Matias, Jr., Kesari Mishra, Yan Qu, Dharmaraja Selvamuthu, Vibhu Sharma, Harish Sukhwani, Alex Thomasian and (late) Ranjith Vasireddy.

Kishor S. Trivedi

Preface to the Second Edition

Nearly 20 years have passed since the publication of the first edition of this book. Its Indian edition is still in print and doing quite well. In this second edition, I have thoroughly revised all the chapters. Many examples and problems are updated, and many new examples and problems have been added. There is a considerable addition of examples on system availability modeling, wireless system performance and availability modeling, software reliability modeling, and system performability modeling. New material on fault trees and stochastic Petri nets, and numerical solution techniques for Markov chains has been added. A section on the computation of response time distribution for Markovian queuing networks has also been added. Chapter 8, on continuous-time Markov chains, has undergone the most change. My research experience and the application of these methods in practice for the past 25 years (at the time of writing) have been distilled in these chapters as much as possible. I hope that the book will be of use as a classroom textbook as well as of use for practicing engineers. Researchers will also find valuable material here. I have tried to avoid adding excessive and very advanced material. Thus, for instance, I have omitted discussion of Markov regenerative processes, fluid stochastic Petri nets, and binary decision diagrams. Other topics that are omitted include material on self-similarity, large deviation theory, and diffusion approximation. The topic of hierarchical and fixed-point iterative models is covered very briefly. Modeling software fault tolerance with various kinds of correlation is also excluded.

I wish to thank many of my current students—Yonghuan Cao, Dong Chen, Dongyan Chen, Christophe Hirel, Lei Li, Yun Liu, Rajiv Poonamalli, Srinivasan Ramani, Kalyan Vaidyanathan, Wei Xie, and Liang Yin; current postdoctoral associates—Dr. Katerina Goseva-Popstojanova, Dr. Yiguang Hong, Dr. Xiaomin Ma, and Dr. Dharmaraja Selvamuthu; former students—Dr. Hoon Choi, Dr. Gianfranco Ciardo, Dr. Ricardo Fricks, Dr. Sachin Garg, Dr. Swapna Gokhale, Wei Li, Xuemei (Judith) Lou, Dr. Tong Luo, Dr. Yue Ma, Dr. Varsha Mainkar, Dr. Manish Malhotra, Anu Mohan, Dr. Jogesh Muppala, Hu Pan, Dr. Anapathur Ramesh, Dr. Robin Sahner, Kartik Sudeep, Dr. Steve Woollet, and Dr. Xinyu (Henry) Zang;

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Kishor S. Trivedi

Preface to the First Edition

The aim of this book is to provide an introduction to probability, stochastic processes, and statistics for students of computer science, electrical/computer engineering, reliability engineering, and applied mathematics. The prerequisites are two semesters of calculus, a course on introduction to computer programming, and preferably, a course on computer organization.

I have found that the material in the book can be covered in a two-semester or three-quarter course. However, through a choice of topics, shorter courses can also be organized. I have taught the material in this book to seniors and first-year graduate students but with the text in printed form, it could be given to juniors as well.

With a specific audience in mind, I have attempted to provide examples and problems, with which the student can identify, as motivation for the probability concepts. The majority of applications are drawn from reliability analysis and performance analysis of computer systems and from probabilistic analysis of algorithms. Although there are many good texts on each of these application areas, I felt the need for a text that treats them in a balanced fashion.

Chapters 1–5 provide an introduction to probability theory. These five chapters provide the core for one semester course on introduction to applied probability. Chapters 6–9 deal with stochastic processes and their applications. These four chapters form the core of the second course with a title such as systems modeling. I have included an entire chapter on networks of queues. The last two chapters are on statistical inference and regression, respectively. I have placed the material on sampling distributions in Chapter 3 dealing with continuous random variables. Portions of the chapters on statistics can be taught with the first course and other portions in the second course.

Besides more than 200 worked examples, most sections conclude with a number of exercises. Difficult exercises are indicated by a star. A solution manual for instructors is available from the publisher.

I am indebted to the Department of Computer Science, Duke University and to Merrell Patrick for their encouragement and support during this project. The efficient typing skills of Patricia Land helped make the job of writing the book much easier than it could have been.

Many of my friends, colleagues, and students carefully read several drafts and suggested many changes that improved the readability and the accuracy of

this text. Many thanks to Robert Geist, Narayan Bhat, Satish Tripathi, John Meyer, Frank Harrell, Veena Adlakha, and Jack Stiffler for their suggestions. Joey de la Cruz and Nelson Strothers helped in the editing and the typing process very early in the project. The help by the staff of Prentice-Hall in preparation of the book is also appreciated.

I would like to thank my wife Kalpana, and my daughters Kavita and Smita for enduring my preoccupation with this work for so long. The book is dedicated to my parents.

Kishor S. Trivedi

Acronyms

APS	automatic protection switching
ARQ	automatic repeat request
ATM	asynchronous transfer mode
BCC	blocked calls cleared
BDD	binary decision diagram
BER	bit error rate
BR	base radio
BTS	base-transceiver system
CDF	cumulative distribution function
CFR	constant failure rate
CME	conditional mean exceedance
CPU	central processing unit
CRTD	conditional response time distribution
CSM	central server model
CTMC	continuous-time Markov chain
DFR	decreasing failure rate
DTMC	discrete-time Markov chain
ESS	electronic switching system
FCFS	first come, first served
FIFO	first in, first out
FIT	failure in time
FTREE	fault tree
GBN	Go Back N
GMR	general(ized) multiplication rule
GSPN	generalized stochastic Petri net
IBP	interrupted Bernoulli process
IFR	increasing failure rate
I/O	input output
IRM	independent reference model
IS	infinite server
ISP	Internet service provider
LAN	local area network
LCFS	last come, first served
LCFS-PR	LCFS-preemptive resume
LCL	lower confidence limit
LLC	logical link control

LRU	least recently used
LST	Laplace–Stieltjes transform
LT	Laplace transform
MAC	medi(um)(a) access control
MAP	Markovian arrival process
METF	mean effort to (security) failure
MGF	moment generating function
MLE	maximum-likelihood estimator
MMBP	Markov modulated Bernoulli process
MMPP	Markov modulated Poisson process
MRM	Markov reward model
MTBF	mean time between failures
MTTF	mean time to failure
MTTR	mean time to repair
MVA	mean-value analysis
NHCTMC	Nonhomogeneous CTMC
NHPP	Nonhomogeneous Poisson process
NPFQN	Non-product-form queuing network(s)
ODE	ordinary differential equation
PASTA	Poisson arrivals see time averages
PN	Petri net
pdf	probability density function
PFQN	Product-form queuing network(s)
pmf	probability mass function
prs	preemptive resume
prt	preemptive repeat
PS	processor sharing
QoS	quality of service
RBD	reliability block diagram
RG	reliability graph
RR	round robin
SDP	sum of disjoint products
SEN	shuffle exchange network
SHARPE	symbolic hierarchical automated reliability and performance evaluator
SLTF	shortest latency time first
SMP	semi-Markov process
SOR	successive overrelaxation
SPN	stochastic Petri net
SPNP	stochastic Petri net package
SREPT	software reliability estimation and prediction tool
SRGM	software reliability growth model
SRM	Software reliability model
SRN	stochastic reward net
SRPTF	shortest remaining processing time first

TDMA	time division multiple access
TLP	total loss probability
TMR	triple modular redundancy
UBT	upside-down bathtub
UCL	upper confidence limit
URTD	unconditional response time distribution
VLSI	very large scale integrated (circuits)
WFS	workstation–file server (system)

About the Companion Website

This book is accompanied by a password protected companion website for instructors only.

This website contains:

- Course Slides
- Solutions Manual

To access the material on the instructor's website simply visit www.wiley.com/go/trivedi/probabilityandstatistics2e and follow the instructions for how to register.

Chapter 1

Introduction

1.1 MOTIVATION

Computer scientists and engineers need powerful techniques to analyze algorithms and computer systems. Similarly, networking engineers need methods to analyze the behavior of protocols, routing algorithms, and congestion in networks. Computer systems and networks are subject to failure, and hence methods for their reliability and availability are needed. Many of the tools necessary for these analyses have their foundations in probability theory. For example, in the analysis of algorithm execution times, it is common to draw a distinction between the *worst-case* and the *average-case* behavior of an algorithm. The distinction is based on the fact that for certain problems, while an algorithm may require an inordinately long time to solve the least favorable instance of the problem, the average solution time is considerably shorter. When many instances of a problem have to be solved, the probabilistic (or average-case) analysis of the algorithm is likely to be more useful. Such an analysis accounts for the fact that the performance of an algorithm is dependent on the distributions of input data items. Of course, we have to specify the relevant probability distributions before the analysis can be carried out. Thus, for instance, while analyzing a sorting algorithm, a common assumption is that every permutation of the input sequence is equally likely to occur.

Similarly, if the storage is dynamically allocated, a probabilistic analysis of the storage requirement is more appropriate than a worst-case analysis. In a like fashion, a worst-case analysis of the accumulation of roundoff errors in a numerical algorithm tends to be rather pessimistic; a probabilistic analysis, although harder, is more useful.

When we consider the analysis of a Web server serving a large number of users, several types of random phenomena need to be accounted for. First, the arrival pattern of requests is subject to randomness due to a large population of diverse users. Second, the resource requirements of requests will likely fluctuate from request to request as well as during the execution of a single request. Finally, the resources of the Web server are subject to random failures due to environmental conditions and aging phenomena. The theory of stochastic (random) processes is very useful in evaluating various measures of system effectiveness such as throughput, response time, reliability, and availability.

Before an algorithm (or protocol) or a system can be analyzed, various probability distributions have to be specified. Where do the distributions come from? We may collect data during the actual operation of the system (or the algorithm). These measurements can be performed by hardware monitors, software monitors, or both. Such data must be analyzed and compressed to obtain the necessary distributions that drive the analytical models discussed above. Mathematical statistics provides us with techniques for this purpose, such as the **design of experiments**, **hypothesis testing**, **estimation**, **analysis of variance**, and **linear and nonlinear regression**.

1.2 PROBABILITY MODELS

Probability theory is concerned with the study of random (or chance) phenomena. Such phenomena are characterized by the fact that their future behavior is not predictable in a deterministic fashion. Nevertheless, such phenomena are usually capable of mathematical descriptions due to certain statistical regularities. This can be accomplished by constructing an idealized probabilistic model of the real-world situation. Such a model consists of a list of all possible outcomes and an assignment of their respective probabilities. The theory of probability then allows us to predict or deduce patterns of future outcomes.

Since a model is an abstraction of the real-world problem, predictions based on the model must be validated against actual measurements collected from the real phenomena. A poor validation may suggest modifications to the original model. The theory of statistics facilitates the process of validation. Statistics is concerned with the inductive process of drawing inferences about the model and its parameters based on the limited information contained in real data.

The role of probability theory is to analyze the behavior of a system or an algorithm assuming the given probability assignments and distributions. The results of this analysis are as good as the underlying assumptions. Statistics helps us in choosing these probability assignments and in the process of validating model assumptions. The behavior of the system (or the algorithm) is observed, and an attempt is made to draw inferences about the underlying unknown distributions of random variables that describe system activity. Methods of statistics, in turn, make heavy use of probability theory.

Consider the problem of predicting the number of request arrivals to a Web server in a fixed time interval $(0, t]$. A common model of this situation is to assume that the number of arrivals in this period has a particular distribution, such as the Poisson distribution (see Chapter 2). Thus we have replaced a complex physical situation by a simple model with a single unknown parameter, namely, the average arrival rate λ . With the help of probability theory we can then deduce the pattern of future arrivals. On the other hand, statistical techniques help us estimate the unknown parameter λ based on actual observations of past arrival patterns. Statistical techniques also allow us to test the validity of the Poisson model.

As another example, consider a fault-tolerant computer system with automatic error recovery capability. Model this situation as follows. The probability of successful recovery is c and probability of an abortive error is $1 - c$. The uncertainty of the physical situation is once again reduced to a simple probability model with a single unknown parameter c . In order to estimate parameter c in this model, we observe N errors out of which n are successfully recovered. A reasonable estimate of c is the relative frequency n/N , since we expect this ratio to converge to c in the limit $N \rightarrow \infty$. Note that this limit is a limit in a probabilistic sense:

$$\lim_{N \rightarrow \infty} P \left(\left| \frac{n}{N} - c \right| > \epsilon \right) = 0.$$

Axiomatic approaches to probability allow us to define such limits in a mathematically consistent fashion (e.g., see the law of large numbers in Chapter 4) and hence allow us to use relative frequencies as estimates of probabilities.

1.3 SAMPLE SPACE

Probability theory is rooted in the real-life situation where a person performs an experiment the outcome of which may not be certain. Such an experiment is called a **random experiment**. Thus, an experiment may consist of the simple process of noting whether a component is functioning properly or has failed; it may consist of determining the execution time of a program; or it may consist of determining the response time of a server request. The result of any such observations, whether they are simple “yes” or “no” answers, meter readings, or whatever, are called **outcomes** of the experiment.

Definition (Sample Space). The totality of the possible outcomes of a random experiment is called the **sample space** of the experiment and it will be denoted by the letter S .

We point out that the sample space is not determined completely by the experiment. It is partially determined by the purpose for which the experiment is carried out. If the status of two components is observed, for some purposes it is sufficient to consider only three possible outcomes: two functioning, two malfunctioning, one functioning and one malfunctioning. These



Figure 1.1. A one-dimensional sample space

three outcomes constitute the sample space S . On the other hand, we might be interested in exactly which of the components has failed, if any has failed. In this case the sample space S must be considered as four possible outcomes where the earlier single outcome of one failed, one functioning is split into two outcomes: first failed, second functioning and first functioning, second failed. Many other sample spaces can be defined if we take into account such things as type of failure and so on.

Frequently, we use a larger sample space than is strictly necessary because it is easier to use; specifically, it is always easier to discard excess information than to recover lost information. For instance, in the preceding illustration, the first sample space might be denoted $S_1 = \{0, 1, 2\}$ (where each number indicates how many components are functioning) and the second sample space might be denoted $S_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ (where 0 = failed, 1 = functioning). Given a selection from S_2 , we can always add the two components to determine the corresponding choice from S_1 ; but, given a choice from S_1 (in particular 1), we cannot necessarily recover the corresponding choice from S_2 .

It is useful to think of the outcomes of an experiment, the **elements** of the sample space, as points in a space of one or more dimensions. For example, if an experiment consists of examining the state of a single component, it may be functioning properly (denoted by the number 1), or it may have failed (denoted by the number 0). The sample space is one-dimensional, as shown in Figure 1.1. If a system consists of two components there are four possible outcomes, as shown in the two-dimensional sample space of Figure 1.2. Here each coordinate is 0 or 1 depending on whether the corresponding component is functioning properly or has failed. In general, if a system has n components, there are 2^n possible outcomes each of which can be regarded as a point in an

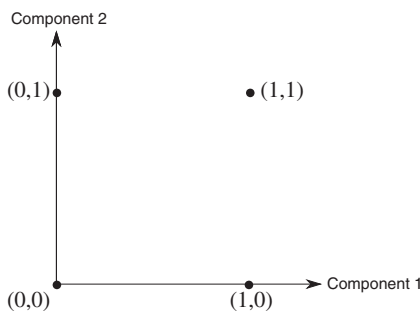


Figure 1.2. A two-dimensional sample space

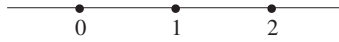


Figure 1.3. A one-dimensional sample space

n -dimensional sample space. It should be noted that the sample space used here in connection with the observation of the status of components could also serve to describe the results of other experiments; for example, the experiment of observing n successive executions of an **if** statement, with 1 denoting the execution of the **then** clause and 0 denoting the execution of the **else** clause.

The geometric configuration that is used to represent the outcomes of an experiment (e.g., Figure 1.2) is not necessarily unique. For example, we could have regarded the outcomes of the experiment of observing the two-component system to be the total number functioning, and the outcomes would be 0,1,2, as depicted in the one-dimensional sample space of Figure 1.3. Note that point 1 in Figure 1.3 corresponds to points (0,1) and (1,0) in Figure 1.2. It is often easier to use sample spaces whose elements cannot be further “subdivided”; that is, the individual elements of a sample space should not represent two or more outcomes that are distinguishable in some fashion. Thus, sample spaces like those of Figures 1.1 and 1.2 should be used in preference to sample spaces like the one in Figure 1.3.

It is convenient to classify sample spaces according to the number of elements they contain. If the set of all possible outcomes of the experiment is finite, then the associated sample space is a **finite sample space**. Thus, the sample spaces of Figures 1.1–1.3 are finite sample spaces.

To consider an example where a finite sample space does not suffice, suppose we inspect components coming out of an assembly line and that we are interested in the number inspected before we observe the first defective component. It could be the first, the second, ..., the hundredth, ..., and, for all we know, we might have to inspect a billion or more before we find a defective component. Since the number of components to be inspected before the first defective one is found is not known in advance, it is appropriate to take the sample space to be the set of natural numbers. The same sample space results for the experiment of tossing a coin until a head is observed. A sample space such as this, where the set of all outcomes can be put into a one-to-one correspondence with the natural numbers, is said to be **countably infinite**. Usually it is not necessary to distinguish between finite and countably infinite sample spaces. Therefore, if a sample space is either finite or countably infinite, we say that it is a **countable** or a **discrete sample space**.

Measurement of the time until failure of a component would have an entire interval of real numbers as possible values. Since the interval of real numbers cannot be enumerated—that is, they cannot be put into one-to-one correspondence with natural numbers—such a sample space is said to be **uncountable** or **nondenumerable**. If the elements (points) of a sample space constitute a continuum, such as all the points on a line, all the points on a line segment or

all the points in a plane, the sample space is said to be **continuous**. Certainly, no real experiment conducted using real measuring devices can ever yield such a continuum of outcomes, since there is a limit to the fineness to which any instrument can measure. However, such a sample space can often be taken as an idealization of, an approximation to, or a model of a real world situation, which may be easier to analyze than a more exact model.

Problems

1. Problems Describe a possible sample space for each of the following experiments:
 - (a) A large lot of RAM (random access memory) chips is known to contain a small number of ROM (read-only memory) chips. Three chips are chosen at random from this lot and each is checked to see whether it is a ROM or a RAM.
 - (b) A box of 10 chips is known to contain one defective and nine good chips. Three chips are chosen at random from the box and tested.
 - (c) An **if...then...else...** statement is executed 4 times.

1.4 EVENTS

An **event** is simply a collection of certain sample points, that is, a subset of the sample space. Equivalently, any statement of conditions that defines this subset is called an event. Intuitively, an event is defined as a statement whose truth or falsity is determined after the experiment. The set of all experimental outcomes (sample points) for which the statement is true defines the subset of the sample space corresponding to the event. A single performance of the experiment is known as a **trial**. Let E be an event defined on a sample space S ; that is, E is a subset of S . Let the outcome of a specific trial be denoted by s , an element of S . If s is an element of E , then we say that the event E has occurred. Only one outcome s in S can occur on any trial. However, every event that includes s will occur.

Consider the experiment of observing a two-component system and the corresponding sample space of Figure 1.2. Let event A_1 be described by the statement “Exactly one component has failed.” Then it corresponds to the subset $\{(0,1), (1,0)\}$ of the sample space. We will use the term **event** interchangeably to describe the subset or the statement. There are sixteen different subsets of this sample space with four elements, and each of these subsets defines an event. In particular, the entire sample space $S = \{(0,0), (0,1), (1,0), (1,1)\}$ is an event (called the **universal event**), and so is the null set $\emptyset$ (called the **null or impossible event**). The event $\{s\}$ consisting of a single sample point will be called an **elementary event**.

Consider the experiment of observing the time to failure of a component. The sample space, in this case, may be thought of as the set of all nonnegative real numbers, or the interval $[0, \infty) = \{t \mid 0 \leq t < \infty\}$. Note that this is an

TABLE 1.1. Sample Points

$s_0 = (0, 0, 0, 0, 0)$	$s_{16} = (1, 0, 0, 0, 0)$
$s_1 = (0, 0, 0, 0, 1)$	$s_{17} = (1, 0, 0, 0, 1)$
$s_2 = (0, 0, 0, 1, 0)$	$s_{18} = (1, 0, 0, 1, 0)$
$s_3 = (0, 0, 0, 1, 1)$	$s_{19} = (1, 0, 0, 1, 1)$
$s_4 = (0, 0, 1, 0, 0)$	$s_{20} = (1, 0, 1, 0, 0)$
$s_5 = (0, 0, 1, 0, 1)$	$s_{21} = (1, 0, 1, 0, 1)$
$s_6 = (0, 0, 1, 1, 0)$	$s_{22} = (1, 0, 1, 1, 0)$
$s_7 = (0, 0, 1, 1, 1)$	$s_{23} = (1, 0, 1, 1, 1)$
$s_8 = (0, 1, 0, 0, 0)$	$s_{24} = (1, 1, 0, 0, 0)$
$s_9 = (0, 1, 0, 0, 1)$	$s_{25} = (1, 1, 0, 0, 1)$
$s_{10} = (0, 1, 0, 1, 0)$	$s_{26} = (1, 1, 0, 1, 0)$
$s_{11} = (0, 1, 0, 1, 1)$	$s_{27} = (1, 1, 0, 1, 1)$
$s_{12} = (0, 1, 1, 0, 0)$	$s_{28} = (1, 1, 1, 0, 0)$
$s_{13} = (0, 1, 1, 0, 1)$	$s_{29} = (1, 1, 1, 0, 1)$
$s_{14} = (0, 1, 1, 1, 0)$	$s_{30} = (1, 1, 1, 1, 0)$
$s_{15} = (0, 1, 1, 1, 1)$	$s_{31} = (1, 1, 1, 1, 1)$

example of a continuous sample space. Now if this component is part of a system that is required to carry out a mission of certain duration t , then an event of interest is “The component does not fail before time t .” This event may also be denoted by the set $\{x \mid x \geq t\}$, or by the interval $[t, \infty)$.

1.5 ALGEBRA OF EVENTS

Consider an example of a wireless cell with five identical channels. One possible random experiment consists of checking the system to see how many channels are currently available. Each channel is in one of two states: busy (labeled 0) and available (labeled 1). An outcome of the experiment (a point in the sample space) can be denoted by a 5-tuple of 0s and 1s. A 0 in position i of the 5-tuple indicates that channel i is busy and a 1 indicates that it is available. The sample space S has $2^5 = 32$ sample points, as shown in Table 1.1. The event E_1 described by the statement “At least four channels are available” is given by

$$\begin{aligned}
 E_1 &= \{(0, 1, 1, 1, 1), (1, 0, 1, 1, 1), (1, 1, 0, 1, 1), (1, 1, 1, 0, 1), \\
 &\quad (1, 1, 1, 1, 0), (1, 1, 1, 1, 1)\} \\
 &= \{s_{15}, s_{23}, s_{27}, s_{29}, s_{30}, s_{31}\}.
 \end{aligned}$$

The **complement** of this event, denoted by $\overline{E}_1$, is defined to be $S - E_1$, and contains all of the sample points not contained in E_1 ; that is, $\overline{E}_1 = \{s \in S \mid s \notin E_1\}$. In our example, $\overline{E}_1 = \{s_0 \text{ through } s_{14},$

s_{16} through s_{22} , s_{24} through s_{26} , s_{28} . $\overline{E}_1$ may also be described by the statement “at most three channels are available.” Let E_2 be the event “at most four channels are available.” Then $E_2 = \{s_0 \text{ through } s_{30}\}$. The **intersection** E_3 of the two events E_1 and E_2 is denoted by $E_1 \cap E_2$ and is given by:

$$\begin{aligned} E_3 &= E_1 \cap E_2 \\ &= \{s \in S \mid s \text{ is an element of both } E_1 \text{ and } E_2\} \\ &= \{s \in S \mid s \in E_1 \text{ and } s \in E_2\} \\ &= \{s_{15}, s_{23}, s_{27}, s_{29}, s_{30}\}. \end{aligned}$$

Let E_4 be the event “channel 1 is available.” Then $E_4 = \{s_{16} \text{ through } s_{31}\}$. The **union** E_5 of the two events E_1 and E_4 is denoted by $E_1 \cup E_4$ and is given by:

$$\begin{aligned} E_5 &= E_1 \cup E_4 \\ &= \{s \in S \mid \text{either } s \in E_1 \text{ or } s \in E_4 \text{ or both}\} \\ &= \{s_{15} \text{ through } s_{31}\}. \end{aligned}$$

Note that E_1 has 6 points, E_4 has 16 points, and E_5 has 17 points. In general:

$$\begin{aligned} |E_5| &= |E_1 \cup E_4| \\ &\leq |E_1| + |E_4|. \end{aligned}$$

Here, the notation $|A|$ is used to denote the number of elements in the set A (also known as the **cardinality** of A).

Two events A and B are said to be **mutually exclusive events** or **dis-joint events** provided $A \cap B$ is the null set. If A and B are mutually exclusive, then it is not possible for both events to occur on the same trial. For example, let E_6 be the event “channel 1 is busy.” Then E_4 and E_6 are mutually exclusive events since $E_4 \cap E_6 = \emptyset$.

Although the definitions of union and intersection are given for two events, we observe that they extend to any finite number of sets. However, it is customary to use a more compact notation. Thus we define

$$\begin{aligned} \bigcup_{i=1}^n E_i &= E_1 \cup E_2 \cup E_3 \cdots \cup E_n \\ &= \{s \text{ element of } S \mid s \text{ element of } E_1 \text{ or } s \text{ element of } E_2 \\ &\quad \text{or } \cdots s \text{ element of } E_n\} \\ \bigcap_{i=1}^n E_i &= E_1 \cap E_2 \cap E_3 \cap \cdots \cap E_n \\ &= \{s \text{ element of } S \mid s \text{ element of } E_1 \text{ and } s \text{ element of } E_2 \\ &\quad \text{and } \cdots s \text{ element of } E_n\} \end{aligned}$$

These definitions can also be extended to the union and intersection of a countably infinite number of sets.

The algebra of events may be fully defined by the following five laws or axioms, where A , B , and C are arbitrary sets (or events), and S is the universal set (or event):

(E1) *Commutative laws:*

$$A \cup B = B \cup A, \quad A \cap B = B \cap A.$$

(E2) *Associative laws:*

$$A \cup (B \cap C) = (A \cup B) \cap C,$$

$$A \cap (B \cup C) = (A \cap B) \cup C.$$

(E3) *Distributive laws:*

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

(E4) *Identity laws:*

$$A \cup \emptyset = A, \quad A \cap S = A.$$

(E5) *Complementation laws:*

$$A \cup \overline{A} = S, \quad A \cap \overline{A} = \emptyset.$$

Any relation that is valid in the algebra of events can be proved by using these axioms [(E1–E5)]. Some other useful relations are as follows:

(R1) *Idempotent laws:*

$$A \cup A = A, \quad A \cap A = A.$$

(R2) *Domination laws:*

$$A \cup S = S, \quad A \cap \emptyset = \emptyset.$$

(R3) *Absorption laws:*

$$A \cap (A \cup B) = A, \quad A \cup (A \cap B) = A.$$

(R4) *de Morgan's laws:*

$$\overline{(A \cup B)} = \overline{A} \cap \overline{B}, \quad \overline{(A \cap B)} = \overline{A} \cup \overline{B}.$$

$$(R5) \quad \overline{(\overline{A})} = A.$$

$$(R6) \quad A \cup (\overline{A} \cap B) = A \cup B.$$

From the complementation laws, we note that A and $\overline{A}$ are mutually exclusive since $A \cap \overline{A} = \emptyset$. In addition, A and $\overline{A}$ are collectively exhaustive since any point s (an element of S) is either in $\overline{A}$ or in A . These two notions can be generalized to a list of events.

A list of events $A_1, A_2, \dots, A_n$ is said to be composed of **mutually exclusive** events if and only if

$$A_i \cap A_j = \begin{cases} A_i & \text{if } i = j, \\ \emptyset & \text{otherwise.} \end{cases}$$

Intuitively, a list of events is composed of mutually exclusive events if there is no point in the sample space that is included in more than one event in the list.

A list of events $A_1, A_2, \dots, A_n$ is said to be **collectively exhaustive** if and only if

$$A_1 \cup A_2 \cdots \cup A_n = S.$$

Given a list of events that is collectively exhaustive, each point in the sample space is included in at least one event in the list. An arbitrary list of events may be mutually exclusive, collectively exhaustive, both, or neither. For each point s in the sample space S , we may define an event $A_s = \{s\}$. The resulting list of events is mutually exclusive and collectively exhaustive (such a list of events is also called a **partition** of the sample space S). Thus, a sample space may be defined as the mutually exclusive and collectively exhaustive listing of all possible outcomes of an experiment.

Problems

- Four components are inspected and three events are defined as follows:

A = “all four components are found defective.”

B = “exactly two components are found to be in proper working order.”

C = “at most three components are found to be defective.”

Interpret the following events:

- $B \cup C$.
- $B \cap C$.
- $A \cup C$.
- $A \cap C$.

- Use axioms of the algebra of events to prove the relations:

- $A \cup A = A$.

- (b) $A \cup S = S$.
- (c) $A \cap \emptyset = \emptyset$.
- (d) $A \cap (A \cup B) = A$.
- (e) $A \cup (\overline{A} \cap B) = A \cup B$.

1.6 GRAPHICAL METHODS OF REPRESENTING EVENTS

Venn diagrams often provide a convenient means of ascertaining relations between events of interest. Thus, for a given sample space S and the two events A and B , we have the Venn diagram shown in Figure 1.4. In this figure, the set of all points in the sample space is symbolically denoted by the ones within the rectangle. The events A and B are represented by certain regions in S .

The union of two events A and B is represented by the set of points lying in either A or B . The union of two mutually exclusive events A and B is represented by the shaded region in Figure 1.5. On the other hand, if A and B are not mutually exclusive, they might be represented by a Venn diagram like Figure 1.6. $A \cup B$ is represented by the shaded region; a portion of this shaded region is $A \cap B$ and is so labeled.

For an event A , the complement $\overline{A}$ consists of all points in S that do not belong to A , thus $\overline{A}$ is represented by the unshaded region in Figure 1.7. The usefulness of Venn diagrams becomes apparent when we see that the following laws of event algebra, discussed in the last section, are easily seen to hold true by reference to Figures 1.6 and 1.7:

$$\begin{aligned}
 A \cap S &= A, \\
 A \cup S &= S, \\
 \overline{(\overline{A})} &= A, \\
 \overline{(A \cup B)} &= \overline{A} \cap \overline{B}, \\
 \overline{(A \cap B)} &= \overline{A} \cup \overline{B}.
 \end{aligned}$$

Another useful graphical device is the **tree diagram**. As an example, consider the experiment of observing two successive executions of an **if** statement in a certain program. The outcome of the first execution of the **if** statement

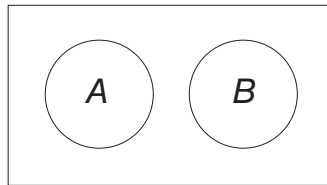


Figure 1.4. Venn diagram for sample space S and events A and B

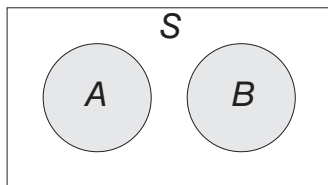


Figure 1.5. Venn diagram of disjoint events A and B

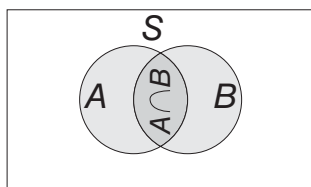


Figure 1.6. Venn diagram for two intersecting events A and B

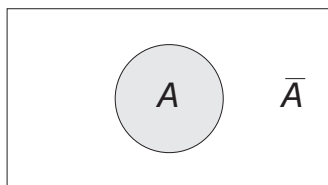


Figure 1.7. Venn diagram of A and its complement

may be the execution of the **then** clause (denoted by T_1) or the execution of the **else** clause (denoted by E_1). Similarly the outcome of the second execution is T_2 or E_2 . This is an example of a **sequential sample space** and leads to the tree diagram of Figure 1.8. We picture the experiment proceeding sequentially downward from the root. The set of all leaves of the tree is the sample space of interest. Each sample point represents the event corresponding to the intersection of all events encountered in tracing a path from the root to the leaf corresponding to the sample point. Note that the four sample points (the leaves of the tree) and their labels constitute the sample space of the experiment. However, when we deal with a sequential sample space, we normally picture the entire generating tree as well as the resulting sample space.

When the outcomes of the experiment may be expressed numerically, yet another graphical device is a coordinate system. As an example, consider a system consisting of two subsystems. The first subsystem consists of four components and the second subsystem contains three components. Assuming that we are concerned only with the total number of defective components in each

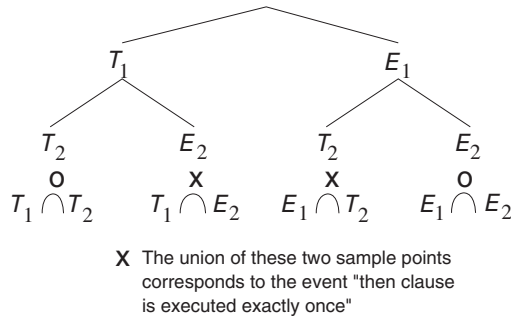


Figure 1.8. Tree diagram of a sequential sample space

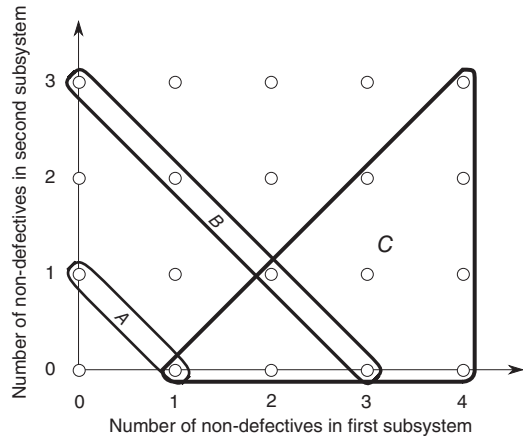


Figure 1.9. A two-dimensional sample space

subsystem (not with what particular components have failed), the cardinality of the sample space is $5 \cdot 4 = 20$, and the corresponding two-dimensional sample space is illustrated in Figure 1.9. The three events identified in Figure 1.9 are easily seen to be

- A = “the system has exactly one non-defective component.”
- B = “the system has exactly three non-defective components.”
- C = “the first subsystem has more non-defective components than the second subsystem.”

1.7 PROBABILITY AXIOMS

We have seen that the physical behavior of random experiments can be modeled naturally using the concepts of events in a suitably defined sample space.

To complete our specification of the model, we shall assign **probabilities** to the events in the sample space. The probability of an event is meant to represent the “relative likelihood” that a performance of the experiment will result in the occurrence of that event. $P(A)$ will denote the probability of the event A in the sample space S .

In many engineering applications and in games of chance, the so-called relative frequency interpretation of the probability is utilized. However, such an approach is inadequate for many applications. We would like the mathematical construction of the probability measure to be independent of the intended application. This leads to an *axiomatic* treatment of the theory of probability. The theory of probability starts with the assumption that probabilities can be assigned so as to satisfy the following three basic **axioms of probability**. The assignment of probabilities is perhaps the most difficult aspect of constructing probabilistic models. Assignments are commonly based on intuition, experience, or experimentation. The theory of probability is neutral; it will make predictions regardless of these assignments. However, the results will be strongly affected by the choice of a particular assignment. Therefore if the assignments are inaccurate, the predictions of the model will be misleading and will not reflect the behavior of the “real world” problem being modeled.

Let S be a sample space of a random experiment. We use the notation $P(A)$ for the probability measure associated with event A . If the event A consists of a single sample point s then $P(A) = P(\{s\})$ will be written as $P(s)$. The probability function $P(\cdot)$ must satisfy the following Kolmogorov’s axioms:

(A1) For any event A , $P(A) \geq 0$.

(A2) $P(S) = 1$.

(A3) $P(A \cup B) = P(A) + P(B)$ provided A and B are mutually exclusive events (i.e., when $A \cap B = \emptyset$).

The first axiom states that all probabilities are nonnegative real numbers. The second axiom attributes a probability of unity to the universal event S , thus providing a normalization of the probability measure (the probability of a certain event, an event that must happen, is equal to 1). The third axiom states that the probability function must be additive. These three axioms are easily seen to be consistent with our intuitive ideas of how probabilities behave.

The **principle of mathematical induction** can be used to show [using axiom (A3) as the basis of induction] that for any positive integer n the probability of the union of n mutually exclusive events $A_1, A_2, \dots, A_n$ is equal to the sum of their probabilities:

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i).$$

The three axioms, (A1)–(A3), are adequate if the sample space is finite but to deal with problems with infinite sample spaces, we need to modify axiom A3:

(A3') For any countable sequence of events $A_1, A_2, \dots, A_n, \dots$, that are mutually exclusive (that is, $A_j \cap A_k = \emptyset$ whenever $j \neq k$):

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

All of conventional probability theory follows from the three axioms (A1) through (A3') of probability measure and the 5 axioms (E1)–(E5) of the algebra of events discussed earlier. These eight axioms can be used to show several useful relations:

(Ra) For any event A , $P(\bar{A}) = 1 - P(A)$.

Proof: A and $\bar{A}$ are mutually exclusive, and $S = A \cup \bar{A}$. Then by axioms (A2) and (A3), $1 = P(S) = P(A) + P(\bar{A})$, from which the assertion follows.

(Rb) If $\emptyset$ is the impossible event, then $P(\emptyset) = 0$.

Proof: Observe that $\emptyset = \bar{S}$ so that the result follows from relation (Ra) and axiom (A2).

(Rc) If A and B are any events, not necessarily mutually exclusive, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof: From the Venn diagram of Figure 1.6, we note that $A \cup B = A \cup (\bar{A} \cap B)$ and $B = (A \cap B) \cup (\bar{A} \cap B)$, where the events on the right-hand side are mutually exclusive in each equation. By axiom (A3), we obtain

$$\begin{aligned} P(A \cup B) &= P(A) + P(\bar{A} \cap B) \\ P(B) &= P(A \cap B) + P(\bar{A} \cap B). \end{aligned}$$

The second equation implies $P(\bar{A} \cap B) = P(B) - P(A \cap B)$, which, after substitution in the first equation, yields the desired assertion.

The relation (Rc) can be generalized to a formula similar to the principle of inclusion and exclusion of combinatorial mathematics [LIU 1968]:

(Rd) If $A_1, A_2, \dots, A_n$ are any events, then

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= P(A_1 \cup A_2 \cup \dots \cup A_n) \\ &= \sum_i P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \end{aligned}$$

$$\begin{aligned}
& + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) + \cdots \\
& + (-1)^{n-1} P(A_1 \cap A_2 \cap \cdots \cap A_n),
\end{aligned}$$

where the successive sums are over all possible events, pairs of events, triples of events, and so on.

Proof: We prove this result by induction on the number of events n . The result (Rc) above can serve as the basis of induction. Assume inductively that (Rd) holds for a union of $n - 1$ events. Define the event $B = A_1 \cup A_2 \cup \cdots \cup A_{n-1}$. Then

$$\bigcup_{i=1}^n A_i = B \cup A_n.$$

Using the result (Rc) above, we get

$$\begin{aligned}
P\left(\bigcup_{i=1}^n A_i\right) &= P(B \cup A_n) \\
&= P(B) + P(A_n) - P(B \cap A_n).
\end{aligned} \tag{1.1}$$

Now

$$B \cap A_n = (A_1 \cap A_n) \cup (A_2 \cap A_n) \cup \cdots \cup (A_{n-1} \cap A_n)$$

is a union of $n - 1$ events and hence, using the inductive hypothesis, we get

$$\begin{aligned}
P(B \cap A_n) &= P(A_1 \cap A_n) + P(A_2 \cap A_n) + \cdots + P(A_{n-1} \cap A_n) \\
&\quad - P[(A_1 \cap A_n) \cap (A_2 \cap A_n)] \\
&\quad - P[(A_1 \cap A_n) \cap (A_3 \cap A_n)] \\
&\quad - \cdots \\
&\quad + P[(A_1 \cap A_n) \cap (A_2 \cap A_n) \cap (A_3 \cap A_n)] \\
&\quad + \cdots - \cdots \\
&\quad + (-1)^{n-2} P[(A_1 \cap A_n) \cap (A_2 \cap A_n) \cap \cdots \cap (A_{n-1} \cap A_n)] \\
&= P(A_1 \cap A_n) + P(A_2 \cap A_n) + \cdots + P(A_{n-1} \cap A_n) \\
&\quad - P(A_1 \cap A_2 \cap A_n) - P(A_1 \cap A_3 \cap A_n) - \cdots \\
&\quad + P(A_1 \cap A_2 \cap A_3 \cap A_n) + \cdots \\
&\quad - \cdots \\
&\quad + (-1)^{n-2} P(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_{n-1} \cap A_n).
\end{aligned} \tag{1.2}$$

Also, since $B = A_1 \cup A_2 \cup \cdots \cup A_{n-1}$ is a union of $n - 1$ events, the inductive hypothesis gives

$$P(B) = P(A_1) + P(A_2) + \cdots + P(A_{n-1})$$

$$\begin{aligned}
& -P(A_1 \cap A_2) - P(A_1 \cap A_3) - \cdots \\
& + \cdots \\
& + (-1)^{n-2} P(A_1 \cap A_2 \cap \cdots \cap A_{n-1}).
\end{aligned} \tag{1.3}$$

Substituting (1.2) and (1.3) into (1.1), we obtain the required result.

The relation (Rd) is computationally expensive to use. A computationally simpler formula is the sum of disjoint products (SDP) formula below.

(Re)

$$\begin{aligned}
P\left(\bigcup_{i=1}^n A_i\right) &= P(A_1) + P(\bar{A}_1 \cap A_2) + P(\bar{A}_1 \cap \bar{A}_2 \cap A_3) + \cdots \\
&+ P(\bar{A}_1 \cap \bar{A}_2 \cap \cdots \cap \bar{A}_{n-1} \cap A_n).
\end{aligned} \tag{1.4}$$

The SDP formula is frequently used in reliability computations [LUO 1998]. We leave the proof as an exercise.

To avoid certain mathematical difficulties, we must place restrictions on which subsets of the sample space may be termed *events* to which probabilities can be assigned. In a given problem there will be a particular class of subsets of S that is “measurable” and will be called the “class of events” $\mathcal{F}$. Since we would like to perform the standard set operations on events, it is reasonable to demand that $\mathcal{F}$ be closed under countable unions of events in $\mathcal{F}$ as well as under complementation. A collection of subsets of a given set S that is closed under countable unions and complementation is called a σ field of subsets of S . Now a **probability space** or **probability system** may be defined as a triple $(S, \mathcal{F}, P)$, where S is a set, $\mathcal{F}$ is a σ -field of subsets of S , and P is a probability measure on $\mathcal{F}$ assumed to satisfy axioms (A1)–(A3’).

If the sample space is discrete (finite or countable), then every subset of S can be an event belonging to $\mathcal{F}$. However, in the case that S is uncountable, this is no longer true. For example, let S be the interval $[0, 1]$ and assume the probability assignment $P(a \leq x \leq b) = b - a$ for $0 \leq a \leq b \leq 1$. Then it can be shown that not all possible subsets of S can be assigned a probability in a manner consistent with the three axioms of P . In such cases, the smallest σ field of subsets of S containing all open and closed intervals is usually adopted as the class of events $\mathcal{F}$.

In summary, P is a function with domain $\mathcal{F}$ and range $[0, 1]$, which satisfies the three axioms (A1), (A2), and (A3’). P assigns a number between 0 and 1 to any event in $\mathcal{F}$. In general, $\mathcal{F}$ does not include all possible subsets of S , and the subsets (events) included in $\mathcal{F}$ are called *measurable*. However, for our purposes, every subset of a sample space constructed here can be considered an event having a probability.

We now outline the steps of a basic procedure to be followed in solving problems [GOOD 1977]:

1. *Identify the sample space S .* The sample space S must be chosen so that all of its elements are mutually exclusive and collectively exhaustive, that is, no two elements can occur simultaneously and one element must occur on any trial. Many of the “trick” probability problems are based on some ambiguity in the problem statement or an inexact formulation of the model of a physical situation. The choice of an appropriate sample space resulting from a detailed description of the model, will do much to resolve common difficulties. Since many choices for the sample space are possible, it is advisable to use a sample space whose elements cannot be further “subdivided”—that is, all possible distinguishable outcomes of the experiment should be listed separately.
2. *Assign probabilities.* Assign probabilities to the elements in S . This assignment must be consistent with the axioms (A1), (A2), and (A3). In practice, probabilities are assigned either on the basis of estimates obtained from past experience, or on the basis of a careful analysis of conditions underlying the random experiment, or on the basis of assumptions, such as the common assumption that various outcomes in a finite sample space are equiprobable (equally likely).
3. *Identify the events of interest.* The events of interest, in a practical situation, will be described by statements. These need to be recast as subsets of the sample space. The laws of event algebra (E1)–(E5) and (R1)–(R6) may be used for any simplification. Pictorial devices such as Venn diagrams, tree diagrams, or coordinate system plots may also be used to advantage.
4. *Compute desired probabilities.* Calculate the probabilities of the events of interest using the axioms (A1), (A2), and (A3') and any derived laws such as (Ra), (Rb), (Rc), (Rd), and (Re). It is usually helpful to express the event of interest as a union of mutually exclusive points in the sample space and summing the probabilities of all points included in the union.

Example 1.1

As a simple illustration of this procedure, consider the example of the wireless cell with 5 channels.

Step 1: An appropriate sample space consists of 32 points (see Table 1.1), each represented by a 5-tuple of 0s and 1s. A 0 in position i indicates that channel i is busy and a 1 indicates that it is available.

Step 2: In the absence of detailed knowledge about the system, we assume that each sample point is equally likely. Since there are 32 sample points, we assign a probability of $\frac{1}{32}$ to each, that is, $P(s_0) = P(s_1) = \cdots = P(s_{31}) = \frac{1}{32}$. It is easily seen that this assignment is consistent with the three probability axioms.

Step 3: Assume that we are required to determine the probability that a call is not blocked, given that the conference call needs at least three channels for its execution. The event E of interest, then, is “three or more channels are available.”